Bidirectional Articulated 3D Gaussian Splatting
A Five-Stage Error-State Kalman Estimation Framework

Restored theoretical architecture: free motion, coherence, rigid motion, joint identification, joint-specific refinement

Theory consolidation; corrected masked RGB and foreground-valid depth measurement – 20 October 2026

Abstract

We specify a staged probabilistic estimation architecture for articulated scenes reconstructed from synchronized calibrated multiview RGB-D data. Two endpoint-trained 3D Gaussian Splatting (3DGS) populations are processed *independently*: a forward free-motion error-state iterated extended Kalman filter (IEKF), a backward free-motion IEKF, an uncertainty-aware rigid-coherence support inference, a rigid-body SE(3) IEKF, a joint-identification bridge from estimated rigid motion, and a revolute/prismatic joint-specific IEKF. Across all continuous-state stages, the physical observation likelihood, alpha-compositing renderer, and analytic Gaussian-level Jacobian are invariant. Only the admissible state manifold, transition model and chain-rule lift into Gaussian poses change. This framework does not use a raw union of the endpoint populations and does not use GT joints, part labels, or geometry during inference. We distinguish derivations inherited from the supplied theory notes from architecture choices added to formalize the requested intermediate rigid stage and joint-estimation bridge. The 20 October revision specifies exact masked RGB target compositing, invalid background depth, and a deterministic analytical quadratic objective equivalent to averaging known artificial backgrounds; it does not introduce a constrained-QP optimizer or alter any stage state manifold.

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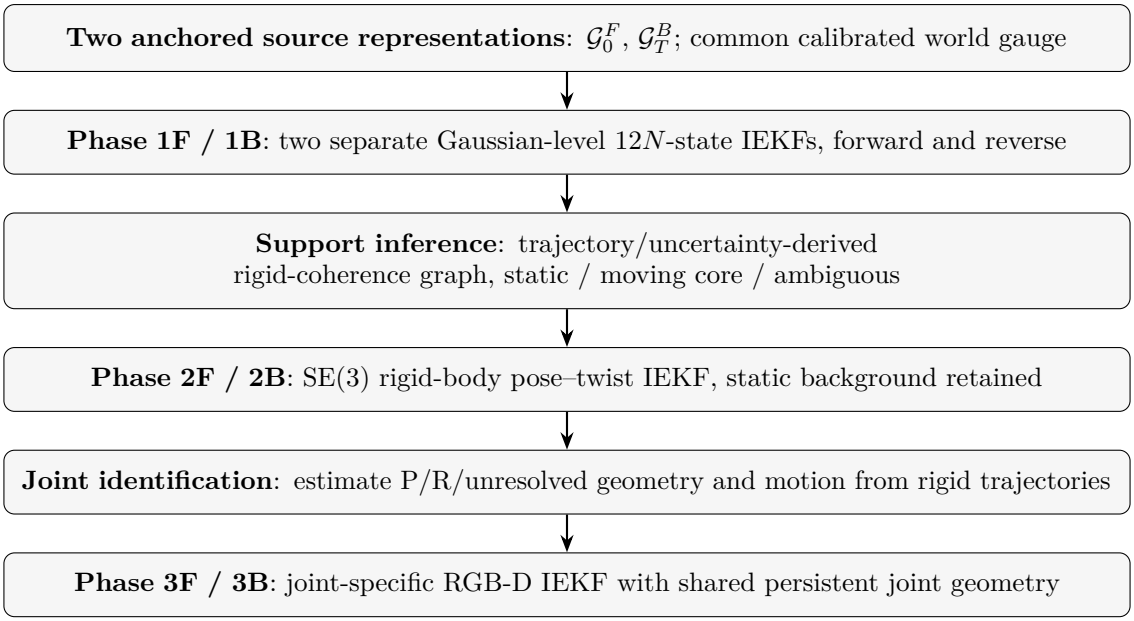
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1 Exact architecture and non-negotiable contracts

Let the measurements be $\mathcal{Y} = \{y_{k,c}\}$ with chronological times $\tau_0 < \cdots < \tau_T$, $h_k = \tau_{k+1} - \tau_k > 0$, and C_k calibrated, *stationary* cameras. Let $\mathcal{G}_0^F$ and $\mathcal{G}_T^B$ be independently trained endpoint 3DGS populations, with source-specific immutable intrinsic Gaussian shapes, opacity and radiance.



Common rules across all stages.

- Raw calibrated RGB, real depth, and supplied whole-object masks define the *prepared* RGB-D observation (Section 13). Depth outside the object or otherwise invalid is never a metric observation. GT part IDs, meshes, joint axes and trajectories can appear only in a post-run *evaluation* process.
- Each source has its own population and own estimator and is anchored at its own training endpoint. Never render $\mathcal{G}^F \cup \mathcal{G}^B$ as though it were one already-valid, nonredundant 3DGS representation.
- Source-trained appearance c_i, opacity o_i and principal shape Λ_i remain fixed in these motion-estimation stages. Means and orientations are transported together; covariances follow exact rigid rotations.
- The *same* explicitly selected image likelihood and differentiable renderer are reused; stage-specific changes occur in the state transition f_s, the state manifold $\mathcal{X}_s$, and the geometry lift $g_s : \mathcal{X}_s \to \mathcal{G}$.
- Phase handoffs carry uncertainty, temporal and identity provenance, and abstention status. A support or joint hypothesis is not converted into ground truth by being passed downstream.

2 Shared physical RGB-D measurement model

A Gaussian i in a given source population has center μ_i, orientation $R_i \in \mathrm{SO}(3)$, principal covariance $\Lambda_i = \mathrm{diag}(\sigma_{i1}^2, \sigma_{i2}^2, \sigma_{i3}^2)$, opacity o_i and radiance c_i. For camera c with calibrated world-to-camera transform $(R_{cw,c}, t_{cw,c})$,

$$m_i = R_{cw,c}\mu_i + t_{cw,c}, \qquad u_i = \pi_c(m_i), \tag{1}$$

$$\Sigma_i = R_i \Lambda_i R_i^\top, \qquad S_{i,c} = J_{\pi,i} R_{cw,c} \Sigma_i R_{cw,c}^\top J_{\pi,i}^\top + \epsilon_s I_2. \tag{2}$$

At image pixel p, contributors are sorted front to back. Within one frozen visibility/sorting branch,

$$\alpha_{p,i} = o_i \exp\left[-\tfrac{1}{2}(p-u_i)^\top S_{i,c}^{-1}(p-u_i)\right], \tag{3}$$

$$T_{p,1} = 1, \qquad T_{p,i+1} = T_{p,i}(1-\alpha_{p,i}), \qquad w_{p,i} = T_{p,i}\alpha_{p,i}, \tag{4}$$

$$\widehat{C}_p = \sum_i w_{p,i} c_i + T_{p,\mathrm{end}} c_{\mathrm{bg}}, \tag{5}$$

$$A_p = \sum_i w_{p,i}, \quad Z_p = \sum_i w_{p,i} d_i, \quad \widehat{D}_p = \frac{Z_p}{A_p + \epsilon_D}. \tag{6}$$

The actual renderer's clamps, appearance basis, masks and depth validity conventions supersede symbolic shorthand. Let $h_c(\mathcal{G})$ stack $\widehat{C}, \widehat{D}$ and let $V_{k,c}$ denote legitimate observed-depth validity. For masked-composite mode, the prepared RGB target and its fixed background must be consistent with the render; observed depth is valid only on real object pixels, as defined in Section 13. The measurement innovation is

$$r_{k,c} = y_{k,c} - h_c(g_s(x_{s,k}; \mathcal{G}^s_{\mathrm{anchor}})), \quad n_{k,c} \sim \mathcal{N}(0, \mathcal{R}_{k,c}). \tag{7}$$

The supplied whole-object segmentation mask may enter observation *preparation*; no ground-truth mesh, part membership or joint state enters Eq. (7). The masked-composite image is a deliberately defined training/estimation target, and artificial background variations are not independent physical sensor measurements.

For one Gaussian let $\delta g_i = (\delta\mu_i, \delta\phi_i) \in \mathbb{R}^6$ be its pose tangent, with left rotational retraction $R_i^+ = \mathrm{Exp}([\delta\phi_i]_\times) R_i$. The existing analytical renderer gives

$$\delta y_{k,c} \simeq H^G_{k,c}\delta g, \qquad H^G_{k,c} = \frac{\partial h_c}{\partial \delta g}; \qquad H_{s,k,c} = H^G_{k,c} J_{g,s,k}, \quad J_{g,s,k} = \frac{\partial g_s}{\partial \delta x_s}. \tag{8}$$

The sparse CUDA JVP/VJP must be the same across stages. RGB/depth couplings and alpha/transmittance interactions remain intact. Velocity and joint-rate image columns are instantaneously zero and become observable through the transition model. The optional background-composite objective in Section 13 is a mathematically specified cost on actual foreground appearance and rendered alpha; it adds no artificial metric-depth observations or hard inequality constraints.

3 Shared IEKF / information-form measurement update

For a prediction $x^- \in \mathcal{X}$ with conditional covariance P^-, use a local manifold retraction $x = x^- \boxplus \delta$. Write $\Lambda^- = (P^-)^{-1}$ on the represented tangent subspace. At inner iteration ℓ, let δ_ℓ be its current tangent coordinate, $r_\ell = y - h(g(x_\ell))$, $H_\ell = H^G_\ell J_{g,\ell}$. For locally Euclidean prior coordinates, the iterated information update is

$$\left(\Lambda^- + H_\ell^\top \mathcal{R}^{-1} H_\ell + \lambda_\ell D_\ell\right) \Delta_\ell = H_\ell^\top \mathcal{R}^{-1} r_\ell - \Lambda^- \delta_\ell, \tag{9}$$

with the appropriate prior-coordinate Jacobian and retraction reset for nonlinear manifolds away from the origin. Candidate steps are accepted using the actual nonlinear, fixed-model MAP objective

$$\mathcal{F}(\delta) = \tfrac{1}{2}\|\delta\|^2_{\Lambda^-} + \tfrac{1}{2}\|y - h(g(x^- \boxplus \delta))\|^2_{\mathcal{R}^{-1}}, \tag{10}$$

(or its exactly specified robust ρ likelihood), evaluating the *same fully specified nonlinear* prepared RGB-D plus analytical contrast cost from Section 13 when that mode is enabled, with a fresh renderer prediction at each trial state. An IEKF inner iteration is a *relinearization of the same observation*, not an additional independent observation. Use Hv and $H^\top z$ matrix free:

$$Au = \Lambda^- u + J_g^\top (H^G)^\top \mathcal{R}^{-1} H^G J_g u. \tag{11}$$

This formula is the shared measurement-information backbone of Phases 1, 2 and 3. The uncertainty $\mathcal{R}$ and process noise must be specified in physical units and consistently transformed into tangent coordinates.

4 Phase 1: two independent Gaussian-level free-motion IEKFs

4.1 State, dynamics and source anchors

For source $s \in \{F, B\}$ with N_s Gaussians, the per-Gaussian state is

$$x^s_{i,k} = (\mu^s_{i,k}, R^s_{i,k}, v^s_{i,k}, \omega^s_{i,k}), \quad \delta x^s_{i,k} = (\delta\mu, \delta\phi, \delta v, \delta\omega) \in \mathbb{R}^{12}. \tag{12}$$

For positive physical time step h_k, the neutral constant-twist transition is

$$\mu^-_{i,k+1} = \mu^+_{i,k} + h_k v^+_{i,k}, \qquad R^-_{i,k+1} = \mathrm{Exp}(h_k[\omega^+_{i,k}]_\times) R^+_{i,k}, \tag{13}$$

$$v^-_{i,k+1} = v^+_{i,k}, \qquad \omega^-_{i,k+1} = \omega^+_{i,k}. \tag{14}$$

The linearized transition uses

$$F_i(h) = \begin{bmatrix} I & 0 & hI & 0 \\ 0 & A & 0 & hJ_\ell(h\omega) \\ 0 & 0 & I & 0 \\ 0 & 0 & 0 & I \end{bmatrix}, \qquad A = \mathrm{Exp}(h[\omega]_\times), \qquad P^-_{k+1} = F_k P^+_k F_k^\top + Q_k. \tag{15}$$

Here J_ℓ is the standard left SO(3) Jacobian, consistently with left-multiplicative rotation errors. The older note wrote J_r in this position; do not propagate that notational inconsistency into the implementation. The exact log-residual Jacobian away from zero differs from the shortcut $[-F, I]$ and must be validated numerically.

Source F starts at the original $\mathcal{G}^F_0$ pose at $k = 0$ and progresses chronologically. Source B starts at $\mathcal{G}^B_T$ at $k = T$ and processes frames in reverse order. A backward filter may store rates with respect to decreasing reverse time; if so, convert to physical chronological rates using $(v, \omega) \mapsto (-v, -\omega)$ and transform cross-covariance blocks accordingly. If velocities are already in physical time, do not flip them.

Endpoint pose constraints: $\mu^F_{i,0} = \bar{\mu}^F_i, R^F_{i,0} = \bar{R}^F_i$ and $\mu^B_{j,T} = \bar{\mu}^B_j, R^B_{j,T} = \bar{R}^B_j$ are declared conditional anchors; velocities remain uncertain. Do not silently anchor backward candidates at $k = 0$. Two independent filters do *not* imply two independent copies of the RGB-D likelihood in a later combined posterior.

4.2 Phase-1 measurement map and uncertainty

For each source separately, $g_1(x)$ simply places every Gaussian at (μ_i, R_i). Therefore

$$H_{1,k,c} = H^G_{k,c} \, \mathrm{diag}_i \begin{bmatrix} I_3 & 0 & 0 & 0 \\ 0 & I_3 & 0 & 0 \end{bmatrix} = \begin{bmatrix} H^\mu_{k,c} & H^\phi_{k,c} & 0 & 0 \end{bmatrix}. \tag{16}$$

The framewise IEKF is Eq. (9) with $x = x^s_k$, $f = f_1$ in Eq. (14), and the source's own $\mathcal{G}^s$. Cross-Gaussian pixel information arises from shared alpha-compositing rays; the operator is not diagonal by Gaussian. Multi-frame velocity observability follows the transition F: $\sum_j \Phi^\top_{j,k} H_j^\top \mathcal{R}_j^{-1} H_j \Phi_{j,k}$.

Required exports: *all* chronological μ, R, v, ω, per-frame covariance/information summaries, actual Q conventions, innovation/NIS diagnostics, source row IDs, anchor, time/rate convention, and rotation/gauge metadata. Existing center-only historical exports do not establish full 12D states and cannot be relabelled as full IEKF output.

5 Rigid-coherence support: probabilistic inference

The input is the posterior *trajectory* and conditional uncertainty obtained from each Phase-1 IEKF. For a spatial-neighbor pair (i, j) in the same source, a rigid transformation preserves distance:

$$r_{ij,k} = \mu_{j,k} - \mu_{i,k}, \qquad \ell_{ij,k} = \|r_{ij,k}\|, \qquad \bar{\ell}_{ij} = \mathrm{median}_{k \in \mathcal{V}_{ij}} \, \ell_{ij,k}, \tag{17}$$

$$z_{ij,k} = \log \frac{\ell_{ij,k} + \epsilon_\ell}{\bar{\ell}_{ij} + \epsilon_\ell}, \qquad C_{ij}^2 = \frac{\sum_{k \in \mathcal{V}_{ij}} w_{ij,k} z_{ij,k}^2}{\sum_{k \in \mathcal{V}_{ij}} w_{ij,k} + \epsilon}. \tag{18}$$

An instantaneous metric rigidity diagnostic is

$$\dot{\ell}_{ij,k}/\ell_{ij,k} = \frac{r_{ij,k}^\top (v_{j,k} - v_{i,k})}{\|r_{ij,k}\|^2}. \tag{19}$$

Weights must account for visibility and estimated uncertainty; if per-Gaussian covariance cross terms are unavailable, the independence approximation must be disclosed. Small metric strain is evidence of *rigid coherence*, not evidence of *motion*. A stationary connected region is equally rigid. Differentiate motion from static by comparing coherent common-rigid hypotheses against a static hypothesis, using temporal displacement and its uncertainty. Allow spatial supernodes/graph connections only within reliable geometric neighborhoods; supernode mixing can make exact support recovery impossible.

Formally introduce latent support labels $z_i \in \{\text{STATIC}, \text{MOVING_RIGID}, \text{AMBIGUOUS}\}$ and a coherence graph $\mathcal{E}$. A conditional support posterior can be written

$$p(z \mid \widehat{X}_1, \widehat{P}_1, \mathcal{E}) \propto \prod_i \psi_i(z_i; \widehat{X}_1, \widehat{P}_1) \prod_{(i,j) \in \mathcal{E}} \psi_{ij}(z_i, z_j; C_{ij}, \widehat{P}_1). \tag{20}$$

Equation (20) is an inference model over discrete identities *conditioned on the preceding IEKF output*; it is not itself an EKF state update and must not be advertised as one. Preserve component labels, supernode-to-Gaussian membership, posterior/score margins, ambiguous regions and the exact exported IDs. The high-precision moving core can be frozen for Phase 2, but do not silently convert a broad binary graph partition into the rigid core. False-positive motion membership is particularly damaging to rigid pose estimation; unresolved geometry should remain explicitly unresolved.

6 Phase 2: rigid-body SE(3) pose–twist IEKF

6.1 State and exact rigid transport

For each selected connected moving rigid part, use one common pose $T_k = \begin{bmatrix} Q_k & b_k \\ 0 & 1 \end{bmatrix} \in \mathrm{SE}(3)$ and a six-dimensional spatial twist $\xi_k = (v_k, \omega_k) \in \mathbb{R}^6$. The 12D rigid error state is

$$x_{2,k} = (T_k, \xi_k), \qquad \delta x_{2,k} = (\delta\tau_k, \delta\xi_k) \in \mathbb{R}^{12}, \qquad T^+ = \mathrm{Exp}_{\mathrm{SE}(3)}(\widehat{\delta\tau}) T. \tag{21}$$

For a moving Gaussian in this source's canonical reference pose, rigid geometry is imposed *exactly*:

$$\mu_{i,k} = Q_k \bar{\mu}_i + b_k, \quad R_{i,k} = Q_k \bar{R}_i, \quad \Sigma_{i,k} = Q_k \bar{\Sigma}_i Q_k^\top. \tag{22}$$

The static Gaussians remain at canonical poses but *stay in the same renderer* for correct occlusion and alpha compositing. Phase 2 has no prismatic/revolute axis in its state and no free per-Gaussian velocity variables. The source anchor frame imposes $T_s = I$ for a source-relative transform, with appropriate local retraction and uncertainty conditioning.

For left/spatial pose perturbations and $\delta g_i = (\delta\mu_i, \delta\phi_i)$, the exact instantaneous kinematic lift is

$$\delta g_i = J_{i,T} \delta\tau, \qquad J_{i,T} = \begin{bmatrix} I_3 & -[\mu_{i,k}]_\times \\ 0 & I_3 \end{bmatrix}. \tag{23}$$

Therefore the Phase-2 image Jacobian is

$$\boxed{H_{2,k,c} = H_{k,c}^G J_{G,T,k} \begin{bmatrix} I_6 & 0 \end{bmatrix}}, \tag{24}$$

where $J_{G,T}$ stacks Eq. (23) for the selected moving Gaussians, with zero blocks for static Gaussian state dependence. The renderer still renders the static Gaussians, and their opacity still influences the H^G of the moving Gaussians.

6.2 Rigid process and IEKF

Use the neutral spatial constant-twist transition

$$T_{k+1}^- = \mathrm{Exp}_{\mathrm{SE}(3)}(h_k\widehat{\xi}_k)T_k^+, \qquad \xi_{k+1}^- = \xi_k^+, \tag{25}$$

with $P_{k+1}^- = F_{2,k}P_k^+F_{2,k}^\top + Q_{2,k}$. Under left pose/twist coordinates its zero-innovation transition linearization has upper blocks $\mathrm{Ad}_{\mathrm{Exp}(h_k\widehat{\xi}_k)}$ and $h_k J_\ell^{\mathrm{SE}(3)}(h_k\xi_k)$; nonzero manifold process residuals require exact inverse-log Jacobians, not the zero-innovation shortcut. The Phase-2 update is *the same Eq.* (9) with $H = H_2$ from Eq. (24). Multiple cameras update *one* shared body pose, not camera-specific poses. There is no generic Adam optimizer and no post-hoc Procrustes/Kabsch substitution for this RGB-D IEKF stage.

The output is $\{T_k, \xi_k, P_{2,k}\}$ and its source-specific support M, with innovation diagnostics and a body-pose covariance gauge convention. A coherent trajectory with a poor support set is still a conditional estimate; preserve the limitation.

7 Joint identification from rigid-state estimates (between Phases 2 and 3)

This stage selects and initializes physical 1-DoF motion hypotheses from *Phase-2 rigid poses and their uncertainty*, without silently pretending that these poses are independent sensor measurements. It is a Bayesian/manifold registration of *inferred rigid states*, not an additional independent RGB-D likelihood.

For source anchor s and each frame k, let $\widehat{T}_{k|s} = \widehat{T}_k\widehat{T}_s^{-1}$. A 1-DoF hypothesis satisfies

$$T_J(q_k;\vartheta_J)T_J(q_s;\vartheta_J)^{-1} = \mathrm{Exp}_{\mathrm{SE}(3)}((q_k - q_s)\widehat{\xi}_J(\vartheta_J)). \tag{26}$$

For **prismatic** motion (P), $e_P \in S^2$ is constant, $q = s$ in metres, and

$$\xi_P = (e_P, 0), \qquad T_P(s) = \begin{bmatrix} I & se_P \\ 0 & 1 \end{bmatrix}. \tag{27}$$

There is *no observable lateral axis point* for pure translation. For **revolute** motion (R), $e_R \in S^2$, a_R is a point on its physical axis and $q = \theta$ in radians:

$$\xi_R = (-e_R \times a_R,\ e_R), \qquad \mu_i(\theta) = a_R + \mathrm{Exp}(\theta[e_R]_\times)(\bar{\mu}_i - a_R). \tag{28}$$

Fix the axis-point gauge by $e_R^\top(a_R - c_0) = 0$ for canonical part reference center c_0; this does *not* assume the physical axis passes through the center of mass. e_R and a_R are constant across the sequence, while q_k can accelerate, reverse or pause.

The uncertainty-weighted pose residual is

$$e_k^J = \mathrm{Log}_{\mathrm{SE}(3)}\Big(\widehat{T}_{k|s}[T_J(q_k;\vartheta_J)T_J(q_s;\vartheta_J)^{-1}]^{-1}\Big)^\vee, \quad \mathcal{F}_J = \tfrac{1}{2}\|e^J\|^2_{\Omega_2^{-1}} + \mathcal{F}_{\mathrm{dyn}}(q,\dot{q}). \tag{29}$$

The joint state can be estimated by a manifold iterated information update on Eq. (29): $x_J = (q_k, \dot{q}_k, \vartheta_J)$, ϑ_J a *persistent* latent geometry, $H_{J,k}^{\mathrm{pose}} = \partial e_k^J/\partial\delta x_J$. If Phase-2 smoothing induces cross-time correlated errors, Ω_2 must include the relevant blocks or be explicitly approximated and

labelled. **Do not stack Phase-2 pose residuals with the same original RGB-D pixels and call them independent observations.**

Report P, R and *unresolved* hypotheses, model fit normalized in comparable units, posterior geometry information/degeneracy, uncertainty, scale/gauge choices, and source/direction. A fitted low-amplitude rotation around a distant axis may mimic translation; a lower raw residual alone does not demonstrate joint type. Identification initializes Phase 3; it is not the final source of RGB-D evidence.

8 Phase 3: joint-specific RGB-D IEKF

8.1 Two physically distinct state manifolds

After resolving or retaining both plausible joint types, create *one filter per hypothesis*, still with the original RGB-D observations and renderer. The state contains one frame-varying joint coordinate and rate plus *time-invariant geometry*:

$$x^P_{3,k} = (s_k, \dot{s}_k, e_P), \qquad e_P \in S^2, \qquad \dim \delta x^P = 2+2 = 4, \tag{30}$$

$$x^R_{3,k} = (\theta_k, \dot{\theta}_k, e_R, a_R), \qquad e_R \in S^2, \quad e_R^\top (a_R - c_0) = 0, \qquad \dim \delta x^R = 2+4 = 6. \tag{31}$$

For each moving Gaussian the complete rigid splat transform is

$$\text{P:} \quad \mu_{i,k} = \bar{\mu}_i + (s_k - s_s) e_P, \qquad R_{i,k} = \bar{R}_i, \tag{32}$$

$$\text{R:} \quad \mu_{i,k} = a_R + Q_R(\theta_k - \theta_s)(\bar{\mu}_i - a_R), \tag{33}$$

$$R_{i,k} = Q_R(\theta_k - \theta_s)\bar{R}_i, \qquad \Sigma_{i,k} = Q_R \bar{\Sigma}_i Q_R^\top. \tag{34}$$

Here the canonical Gaussian geometry is anchored at $s \in \{0, T\}$, so source-B uses the *endpoint-relative* joint displacement rather than pretending its trained Gaussians live at frame 0.

8.2 Joint-specific measurement Jacobians

The chain-rule bridge is

$$\boxed{H^P_3 = H^G J^P_{G,(s,e)}, \qquad H^R_3 = H^G J^R_{G,(\theta,e,a)}}, \tag{35}$$

with zero instantaneous image columns for $\dot{s}$ and $\dot{\theta}$. Let $B_e \in \mathbb{R}^{3\times 2}$ be a tangent basis for the unit direction. For prismatic transport,

$$\frac{\partial \mu_i}{\partial s} = e_P, \quad \frac{\partial \mu_i}{\partial \delta e} = (s_k - s_s) B_e, \quad \frac{\partial \phi_i}{\partial s} = 0, \quad \frac{\partial \phi_i}{\partial \delta e} = 0. \tag{36}$$

For revolute motion, differentiating physical angle yields

$$\frac{\partial \mu_i}{\partial \theta} = e_R \times (\mu_i - a_R), \qquad \frac{\partial \phi_i}{\partial \theta} = e_R. \tag{37}$$

The derivative with respect to the axis parameters is computed with the analytical SE(3) left Jacobian. Put $\zeta = (\theta_k - \theta_s)\xi_R(\vartheta_J)$; for a geometry tangent $\delta\vartheta_J$,

$$\delta\tau = J^{\mathrm{SE}(3)}_\ell(\zeta)(\theta_k - \theta_s)\frac{\partial \xi_R}{\partial \delta \vartheta_J}\delta\vartheta_J, \qquad \delta g_i = \begin{bmatrix} I & -[\mu_i]_\times \\ 0 & I \end{bmatrix} \delta\tau. \tag{38}$$

The axis-point/direction chart must enforce its normalization and axial gauge, and the tangent must be retracted and covariance-reset after accepted updates. Eq. (38) updates *the same axis parameters for all Gaussians and all frames*; no per-frame axis and no per-Gaussian axis is allowed.

8.3 Joint-specific process, information update and geometric invariance

For either type, the process is a scalar kinematic transition with white acceleration noise and a constant latent joint geometry:

$$\begin{bmatrix} q_{k+1} \\ \dot{q}_{k+1} \end{bmatrix}^- = \begin{bmatrix} 1 & h_k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} q_k \\ \dot{q}_k \end{bmatrix}^+, \quad Q_{q\dot{q}}(h_k) = \sigma_a^2 \begin{bmatrix} h_k^3/3 & h_k^2/2 \\ h_k^2/2 & h_k \end{bmatrix}, \quad \vartheta_{J,k+1}^- = \vartheta_{J,k}^+. \tag{39}$$

$Q_{\vartheta_J} = 0$ represents *physical* time invariance; it does not make the posterior geometry covariance zero. The time-varying coordinate/rate cross-covariance with joint geometry must be retained, and tangent covariance resets must be applied when the manifold chart changes.

For P or R, apply Eq. (9) with $H = H_3^P$ or $H = H_3^R$ from Eq. (35). All cameras contribute to the same joint update

$$\mathcal{I}_{3,k} = \Lambda_{3,k}^- + \sum_{c \in C_k} (H_{3,k,c}^J)^\top \mathcal{R}_{k,c}^{-1} H_{3,k,c}^J. \tag{40}$$

The joint geometry is corrected *directly by the prepared RGB-D innovation* $r_{k,c}$. This is the defining mathematical distinction between Phase 3 and post-hoc least-squares joint reduction from $\{\widehat{T}_k\}$.

9 Inference across endpoints, data reuse and model comparison

The F and B pipelines use the same recorded RGB-D and whole-object masks and are thus statistically *correlated*. Treat their outputs as separate source-conditioned estimates and diagnostics. One may compare their axes and kinematics in the verified common gauge, or later design a separate, valid canonical scene representation, but one may *not* add their information matrices as though two independent experiments occurred.

Likewise, Phases 1, 2 and 3 reuse the same images. A Phase-1 or Phase-2 posterior obtained from those images is not an *independent* Gaussian prior to be multiplied with the same pixel likelihood again. Valid uses at stage transitions include numerical initialization, proposal generation, retained representation of conditioned latent structure with clearly limited uncertainty interpretation, or a principled likelihood-correction formulation. When Phase 3 refits the same images, begin with an explicit defensible prior and use earlier posteriors as warm starts/diagnostics rather than silently claiming an exact independent Bayesian fusion.

For joint type, fit and validate both P and R under the same chosen prepared RGB-D measurement model, accounting for extra geometry degrees of freedom and state uncertainty. A calibrated held-out-camera/time predictive likelihood can help distinguish plausible hypotheses. Where the data do not resolve translation from distant-axis rotation, return "unresolved" rather than inserting an arbitrary threshold. There is no dynamic feedback from GT evaluation to inference.

10 Execution and mathematical validation contracts

1. **Shared renderer and masked data:** verify alpha/transmittance, normalized depth, JVP/VJP adjointness and finite differences against the actual production renderer. Verify matched artificial RGB-background compositing, premultiplied mask resolution, and identically zero depth information outside the real foreground mask; see Section 13.
2. **Phase 1:** verify left-SO(3) transition Jacobians and exact nonzero-residual process Jacobians; check forward/reverse physical-rate conversion and covariance signs; export full $12N$ trajectories (not centers alone).

3. **Support:** reproduce trajectory strain, graph edges, component IDs, static/moving/ambiguous membership *row by row*. Audit mixed-label supernodes using GT only in evaluation; never use GT to repair predicted membership.
4. **Phase 2:** verify $J_{i,T}$ numerically for means and anisotropic covariance orientation; analytic composed H_2 JVP/VJP; framewise nonlinear RGB-D reduction conditioned on the same fixed support.
5. **Joint identification:** verify SE(3) pose convention, axis gauge, uncertainty units, and P/R residual comparability; record unresolved and small-motion degeneracies.
6. **Phase 3:** verify H_3^P and H_3^R by finite differences, including the *joint-geometry columns*; test a synthetic axis-offset revolute joint (not passing through COM) and translation without a fictitious axis point; confirm image innovations modify persistent geometry.
7. **Across stages:** no raw endpoint population union, no GT choice criteria, no double counting of observation likelihood, no coefficient or process covariance changes solely to force convergence.

11 Convergence and identifiability: defensible statements

Within a fixed renderer visibility branch, a fixed support, a fixed joint hypothesis and a local manifold chart, suppose the nonlinear measurement map has locally Lipschitz derivatives, the represented prior and measurement information are positive definite, and an Armijo backtracking rule is used. Then the LM/IEKF step from Eq. (9) is a descent direction for the specified conditional MAP objective; accepted objective values are nonincreasing and bounded below. Under standard boundedness assumptions, accumulation points satisfy first-order stationarity. *This is not global convergence to true geometry.*

Support graph inference is a discrete model-selection stage with potentially different local optima and cannot be assigned IEKF convergence results. The P/R joint family is not uniquely distinguishable without sufficient motion amplitude, visible asymmetry and independent viewing constraints. Occlusion creates unobservable pose directions; low-rank Gaussian orientation is inherently unidentifiable for nearly isotropic splats. The hard endpoint anchor is conditional on trained geometry, whose own estimation uncertainty is not fully represented. The constant per-Gaussian velocity model in Phase 1 does not exactly describe rigid revolute motion; its physically meaningful process innovation is $\mu_{k+1} - \mu_k - h_k v_k$, not raw displacement.

12 Theory-to-implementation stage summary

Stage	State / inference	Prepared RGB-D and EKF specialization
Phase 1 F/B	N_s Gaussian states (μ, R, v, ω), independently anchored	$H_1 = [H^\mu, H^\phi, 0, 0]$; free constant twist; separate forward and backward information-form IEKF
Support	Discrete coherent/static/ambiguous membership, graph posterior	*No new image IEKF*; derives coherent moving core from Phase-1 posterior tracks and uncertainty
Phase 2	Rigid state $(T, \xi) \in \mathrm{SE}(3) \times \mathbb{R}^6$	$H_2 = H^G J_{G,T}$; same renderer and IEKF; exact rigid centers, orientations and covariance transport
Joint identification	P/R/unresolved geometry and scalar trajectory fitted to $(\widehat{T}, P_2)$	Bayesian manifold pose-fitting; initializes Phase 3, does not constitute independent RGB-D evidence
Phase 3	P: $(s, \dot{s}, e)$; R: $(\theta, \dot{\theta}, e, a)$	$H_3 = H^G J_{G,J}$; joint-specific physical motion and static geometry parameters refined by *direct* RGB-D IEKF

Provenance and scope

Source-derived: Original *Sparse Alpha-Aware Error-State IEKF for Dynamic 3D Gaussian Splatting* (12-page supplied theory note) supplies the free Gaussian state, analytic projected-covariance/alpha Jacobians, information-form IEKF, observability and local convergence. The supplied *Rigid-Screw Phase 2* 18-page note supplies rigid joint geometry, exact rigid covariance transport, constant joint geometry and direct RGB-D Jacobian $H_G J_{G,J}$; its older "Phase 2" is renamed *Phase 3* here to accommodate the user's explicitly requested intermediate rigid-pose estimation. The supplied forward/backward fusion revision provides independent endpoint anchoring, physical-time reverse-state conventions and warnings against raw endpoint-union rendering or double use of the same measurements.

Explicit architecture formalization (not asserted verbatim by the PDFs): an intermediate *unconstrained rigid* SE(3) *IEKF* as Phase 2, a separate uncertainty-aware rigid-coherence graph support handoff, a manifold pose-level joint-identification bridge, and independent P/R Phase-3 filters. These are the requested staged architecture, consistent with the common renderer theory, but they are *not* claimed to have been fully derived as this exact five-stage pipeline in any single original PDF.

13 Exact masked RGB-background compositing and foreground-valid depth (20 October revision)

Status of prior addendum. This section *replaces* the 19 October proposal to fit hard negative-ray visibility constraints through a separate constrained quadratic program. No hard-constraint QP, constraint-generation loop, second-order feasibility correction, distance-transform pull or synthetic depth background belongs to the method specified here. The original Gaussian RGB compositing JVP/VJP was verified correct for nonzero constant backgrounds in the 19 October CUDA audit; preserve that code. The present method uses the whole-object masks to construct a well-defined RGB observation and a physically valid depth-precision mask, and optionally the *exact expected quadratic* artificial-background contrast objective. All five staged estimators remain as above.

13.1 What the earlier static experiment actually established

The successful static masked-3DGS fix was Splatfacto-style *transparency carving*: choose a uniform random RGB background b on each optimization step, render over precisely this b, construct a premultiplied foreground RGB target composited over the *same* b, and compare the *full predicted RGB image* with that target. The static training retained its original L1+SSIM objective and trainable Gaussian opacity and geometry. In the controlled 30,000-iteration comparison with identical initialization and view schedule, reported background alpha leakage decreased from 0.4247 to 1.1789×10^{-4} (ratio 3602.1). The 99th-percentile Gaussian anisotropy did not decrease; it went from 4.06×10^5 to 4.27×10^5. Therefore the experiment establishes the effectiveness of transparency supervision for *that static trainable model*, not a solution for anisotropy or an identifiability guarantee for *frozen-opacity* motion IEKF. The exact expected *quadratic* construction below is mathematically equivalent to background averaging under quadratic error; it is *not* asserted to be exactly equivalent to expected static L1+SSIM.

13.2 Immutable measured inputs and exactly defined preparation

For each chronological frame k, calibrated fixed camera c and native-resolution pixel p, denote original observed RGB $I_{kcp} \in \mathbb{R}^3$, measured metric depth d^{obs}_{kcp} with independent original validity flag $V^D_{kcp} \in \{0,1\}$, and a supplied *whole-object* foreground mask $M_{kcp} \in \{0,1\}$. The mask is not a moving-part label, joint annotation or GT mesh. For the intended synthetic object-only data, $M = 0$ means no observed modeled object. Define the premultiplied observed foreground

$$P_{kcp} = M_{kcp} I_{kcp}, \qquad \boxed{V^{D,\text{used}}_{kcp} = V^D_{kcp} \wedge (M_{kcp} = 1)}. \tag{41}$$

Choose a fixed (nonoptimized) RGB background $b \in [0,1]^3$ during each prediction and its matched target. Define

$$\boxed{Y^C_{kcp}(b) = P_{kcp} + (1 - M_{kcp})b}. \tag{42}$$

The depth value outside $V^{D,\text{used}} = 1$ is *undefined as an observation*; a zero stored in a tensor is a placeholder, *not* a target depth of zero. All background-depth precision entries must be *exactly zero*. The same depth validity applies in forwards and backwards and all subsequent stages. The predicted depth remains the actual existing normalized Gaussian depth $\widehat{D} = Z/(A + \epsilon_D)$, with no artificial far-plane or background depth. Do not post-mask the rendered RGB. Post-masking its RGB error deletes foreground opacity evidence from negative pixels.

If RGB/mask images must be resized from native binary data, apply the same area/coverage operation $\mathcal{R}$ to the *premultiplied* original foreground and the mask separately:

$$P' = \mathcal{R}(MI), \qquad M' = \mathcal{R}(M), \qquad Y'(b) = P' + (1 - M')b. \tag{43}$$

This commutes exactly with a linear area-average compositor for constant b and does not create a dark edge halo. After resampling, $M' \in [0,1]$ is fractional coverage; it is *not* a binary depth validity indicator. Derive depth validity from true original sensor support and its declared geometric resampling rule; never threshold fractional mask coverage without defining that rule. At native binary resolution, Eq. (41) is definitive.

13.3 Exact Gaussian RGB, depth and alpha expressions

Within the production CUDA renderer's compositing branch, let $w_{pi} = T_{pi}\alpha_{pi}$ be the ordered foreground Gaussian weights, including the real alpha clamps and transmittance stopping rule, and write

$$F_p(x) = \sum_i w_{pi}(x) c_i(x), \quad A_p(x) = \sum_i w_{pi}(x) = 1 - T_{p,\mathrm{end}}(x), \tag{44}$$

$$Z_p(x) = \sum_i w_{pi}(x) d_i(x), \quad \widehat{D}_p(x) = \frac{Z_p(x)}{A_p(x) + \epsilon_D}, \tag{45}$$

$$\boxed{C_p(x;b) = F_p(x) + (1 - A_p(x))b}. \tag{46}$$

The foreground evaluated colors $c_i(x)$ remain frozen only to the extent justified by the actual renderer's radiance convention; if motion changes their evaluated value, the JVP must include $\sum_i w_i\, \delta c_i$. With a fixed background color, $\delta b = 0$ but $\delta T_{\mathrm{end}} \neq 0$ generally. Therefore

$$\boxed{\delta C_p(x;b) = \delta F_p - b\,\delta A_p, \qquad H_{C,p}(b) = H_{F,p} - bH_{A,p}}. \tag{47}$$

If the color basis is fixed with respect to this perturbation, this also equals $\sum_i (c_i - b)\delta w_{pi}$. No extra background state or background covariance is introduced. The original renderer's JVP/VJP/Fisher background term was already validated numerically: do not remove it or modify it based on the camouflage diagnosis.

For any selected fixed b, the *exact* RGB model error is

$$\boxed{e_p(x;b) = C_p(x;b) - Y_p^C(b) = F_p(x) - P_p + (M_p - A_p(x))b}. \tag{48}$$

This error is evaluated on *every image pixel*. Metric depth is evaluated only over valid foreground pixels:

$$E_D(x) = \frac{1}{2} \sum_{p: V_p^{D,\mathrm{used}}=1} (\widehat{D}_p(x) - d_p^{\mathrm{obs}})^\top W_p^D (\widehat{D}_p(x) - d_p^{\mathrm{obs}}). \tag{49}$$

There is no implied depth measurement when $V^{D,\mathrm{used}} = 0$, and the renderer need not output zero depth there.

13.4 Exact deterministic expectation of the randomized-background *quadratic* cost

Let the artificial RGB background be a single random vector b for an entire rendered image, with *known, fixed* mean $\bar{b} = \mathbb{E}[b]$ and covariance $\Sigma_b = \mathbb{E}[(b - \bar{b})(b - \bar{b})^\top]$. For each pixel, let W_p^C be a declared *fixed* symmetric positive semidefinite RGB precision matrix, independent of artificial b and the state during the quadratic evaluation. Define

$$\bar{e}_p(x) = F_p(x) - P_p + (M_p - A_p(x))\bar{b}, \quad \kappa_p = \mathrm{tr}(W_p^C \Sigma_b). \tag{50}$$

Proposition 1 (Exact elimination of artificial-background sampling)**.** *For every Gaussian state x and every such fixed W_p^C,*

$$\boxed{\mathbb{E}_b\left[\frac{1}{2} e_p(x;b)^\top W_p^C e_p(x;b)\right] = \frac{1}{2}\bar{e}_p(x)^\top W_p^C \bar{e}_p(x) + \frac{1}{2}\kappa_p (A_p(x) - M_p)^2.} \tag{51}$$

Proof. From Eq. (48), $e_p(x;b) = \bar{e}_p(x) + (M_p - A_p(x))(b - \bar{b})$. Expand the quadratic. The cross term has zero expectation by definition of $\bar{b}$. The remaining second moment is $\mathbb{E}[(b-\bar{b})^\top W_p^C (b-\bar{b})] = \operatorname{tr}(W_p^C \Sigma_b)$. There is no approximation. Summing over pixels preserves the equality by linearity of expectation, even though the same b is shared across pixels. □

The static carving sampler used independent $b_r, b_g, b_b \sim \mathcal{U}[0,1]$. Thus

$$\bar{b} = \tfrac{1}{2}\mathbf{1}_3, \qquad \Sigma_b = \tfrac{1}{12} I_3, \qquad \boxed{\kappa_p = \tfrac{1}{12}\operatorname{tr}(W_p^C)}. \tag{52}$$

In particular $W_p^C = \sigma_I^{-2} I_3$ gives $\kappa_p = 1/(4\sigma_I^2)$, and the actual scalar contribution is $\frac{1}{2}\kappa_p (A - M)^2$. This coefficient is *derived*, not hand-tuned. For the same quadratic cost, one can also use exact deterministic cubature: evaluate the eight equally weighted backgrounds with each channel in $\{\frac{1}{2} - 1/\sqrt{12}, \frac{1}{2} + 1/\sqrt{12}\}$. They exactly reproduce first and second moments, but are computationally unnecessary. Two gray backgrounds suffice only when the RGB precision is channel diagonal; otherwise they introduce spurious channel cross-covariance. Equation (51) is preferable.

Scope of the exactness statement. The formula is exact for the specified *quadratic* objective with fixed W_p^C. The static training's L1+SSIM expectation is *not* equal to this expression. Neither is an arbitrary residual-dependent Student-t/IRLS likelihood. The implementation must explicitly select this new quadratic RGB mode, or compute the exact selected nonquadratic objective rather than silently invoking Eq. (51) as though it remained exact. A deterministic quadratic objective is a deliberate estimator specification, not a claim that invented background colors were independent measured photographs.

13.5 Analytic IEKF Jacobian and matrix-free normal equation

For one residual linearization around x_ℓ, define the pose-composed derivatives

$$H_{p,s}^C = (H_{F,p} - \bar{b} H_{A,p}) J_{g,s}, \qquad H_{p,s}^A = H_{A,p} J_{g,s}, \qquad H_{p,s}^D = H_{D,p} J_{g,s}. \tag{53}$$

Here $J_{g,s}$ is the original map for free Gaussian, rigid-body or joint motion; velocities/rates contribute through dynamics, not direct framewise pixels. Let observation-minus-prediction innovations be $r_p^C = -\bar{e}_p$, $r_p^A = M_p - A_p$, and $r_p^D = d_p^{\text{obs}} - \widehat{D}_p$ on valid pixels. The *exact local Gauss–Newton / IEKF normal operator for this objective* is

$$\begin{aligned} \mathcal{N}_s = \Lambda_s^- &+ \sum_p (H_{p,s}^C)^\top W_p^C H_{p,s}^C + \sum_p \kappa_p (H_{p,s}^A)^\top H_{p,s}^A \\ &+ \sum_{p: V_p^{D,\text{used}}=1} (H_{p,s}^D)^\top W_p^D H_{p,s}^D, \end{aligned} \tag{54}$$

$$\begin{aligned} b_s = b_{\text{prior}} &+ \sum_p (H_{p,s}^C)^\top W_p^C r_p^C + \sum_p \kappa_p (H_{p,s}^A)^\top r_p^A \\ &+ \sum_{p: V_p^{D,\text{used}}=1} (H_{p,s}^D)^\top W_p^D r_p^D. \end{aligned} \tag{55}$$

The normal-equation solve remains one IEKF update $\mathcal{N}_s \Delta = b_s$ (with the same LM/trust-region damping and manifold-prior retraction conventions from Eq. (9)). The matrix-free alpha contribution is exactly

$$v \mapsto J_{g,s}^\top H_A^\top K H_A J_{g,s} v, \quad K = \operatorname{diag}_p(\kappa_p), \tag{56}$$

with right-hand contribution $J_{g,s}^\top H_A^\top K (M - A)$. This demands a correct alpha JVP and VJP, *not* a second fictitious RGB camera view. The already validated `_C_vis` full-ray alpha operator may be reused (if its alpha output is consistent with the primal render), and the RGB JVP/VJP stays as it is. Every test of the actual nonlinear candidate uses the *same deterministic cost* in

Eq. (51) plus Eq. (49); one may not optimize the contrast term but omit it in acceptance. The Fisher/Gauss–Newton information is locally exact for the chosen least-squares linearization, not the full second derivative of the nonlinear renderer: an IEKF necessarily relinearizes. Posterior calibration requires a separate justified noise interpretation; a synthetic background distribution is not new independent camera evidence.

13.6 What the method proves, what it cannot prove, and tests

Proposition 2 (Color camouflage removed at zero residual)**.** *Consider two distinct background colors $b_1 \neq b_2$. If the full compositing residual in Eq.* (48) *vanishes for both at a pixel, then $A_p = M_p$ and $F_p = P_p$.*

Proof. Subtract the two residual equations to obtain $(M_p - A_p)(b_1 - b_2) = 0$. At least one component of $b_1 - b_2$ is nonzero, so $A_p = M_p$; substitute back to get $F_p = P_p$. □

Proposition 3 (No universal motion recovery with frozen opacity)**.** *If an isolated translational Gaussian footprint $A(p - u)$ is completely contained in a homogeneous background region of unbounded support, with no other foreground/temporal observation and invariant shape/opacity, then $\int A(p - u)^2\, dp$ is constant in u, and its spatial translation gradient is zero. Therefore even the exact background-composite cost need not uniquely determine a direction restoring a stranded Gaussian to the physical object.*

Proof. Substitute $v = p - u$ in the integral. The domain is translation-invariant and the integrand depends only on v; thus $\partial_u \int A(p - u)^2\, dp = 0$. □

The static carving experiment could reduce trainable Gaussian opacity or change geometry; motion-only IEKF retains the source-trained opacity and intrinsic shape. The proposed mode removes fixed-color camouflage as an *objective degeneracy* but is not a guarantee of true-part assignment, global observability, alpha elimination, removal of original training artifacts or zero negative-ray opacity. Keep the valid endpoint anchor, temporal process and later rigid/joint evidence; report what is unobservable instead of manufacturing distance-transform forces.

Required verification: compare RGB composites of prediction/target for arbitrary backgrounds and fractional mask coverage; verify depth has zero precision and zero adjoint/normal contribution outside $V^{D,\text{used}}$, regardless of a stored zero; verify Eq. (51) with analytical moment calculations and exact cubature; test alpha JVP/VJP/adjoint and all three motion lifts by finite differences; test nonlinear energy acceptance and disabled-mode parity; report full-scene runtime on 8 and 64 cameras. No modification to static training, PLY archives or existing output directories.